微分几何第十四次作业参考解答

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1 教材外补充题目

1. (1) （过程见讲义 Lec27 第 13-14 页）

$$(1+f_y^2)f_{xx}-2f_xf_yf_{xy}+(1+f_x^2)f_{yy}=0$$

(2) 根据下述计算验证即可

正螺面: $f_x=\frac{-y}{x^2+y^2}$, $f_y=\frac{x}{x^2+y^2}$, $f_{xx}=\frac{2xy}{(x^2+y^2)^2}$, $f_{xy}=\frac{-x^2+y^2}{(x^2+y^2)^2}$, $f_{yy}=\frac{-2xy}{(x^2+y^2)^2}$,

悬链面:$f_x=\frac{x}{\sqrt{(x^2+y^2)(x^2+y^2-1)}}$, $f_y=\frac{y}{\sqrt{(x^2+y^2)(x^2+y^2-1)}}$, $f_{xx}=\frac{-x^4+y^4-y^2}{\sqrt{(x^2+y^2)^3(x^2+y^2-1)^3}}$, $f_{xy}=\frac{-xy(2x^2+2y^2-1)}{\sqrt{(x^2+y^2)^3(x^2+y^2-1)^3}}$, $f_{yy}=\frac{x^4-y^4-x^2}{\sqrt{(x^2+y^2)^3(x^2+y^2-1)^3}}$,

Scherk 曲面: $f_x=-\tan x$, $f_y=\tan y$, $f_{xx}=-\frac{1}{\cos^2 x}$, $f_{xy}=0$, $f_{yy}=\frac{1}{\cos^2 y}$. □

2. (1)

$$\frac{\partial[u,v]}{\partial[x,y]}=\begin{pmatrix}1+\varphi_{xx} & \varphi_{xy}\\ \varphi_{xy} & 1+\varphi_{yy}\end{pmatrix}$$

$$\frac{\partial[x,y]}{\partial[u,v]}=\frac{1}{2+\varphi_{xx}+\varphi_{yy}}\begin{pmatrix}1+\varphi_{yy} & -\varphi_{xy}\\ -\varphi_{xy} & 1+\varphi_{xx}\end{pmatrix}$$

故 $Re(F)_u=\frac{1}{2+\varphi_{xx}+\varphi_{yy}}(\varphi_{yy}-\varphi_{xx})$, $Re(F)_v=-2\frac{1}{2+\varphi_{xx}+\varphi_{yy}}\varphi_{xy}$, $Im(F)_u=2\frac{1}{2+\varphi_{xx}+\varphi_{yy}}\varphi_{xy}$, $Im(F)_v=\frac{1}{2+\varphi_{xx}+\varphi_{yy}}(\varphi_{yy}-\varphi_{xx})$, 可得 $F(u+iv)$ 是复解析函数。

$$F'=\frac{1}{2+\varphi_{xx}+\varphi_{yy}}(1+\varphi_{yy}-\varphi_{xx}-2i\varphi_{xy}),$$

在 $\varphi_{xx}>0$ 时，$\varphi_{yy}>0, |Re(F)_u| \leqslant \frac{1}{2+\varphi_{xx}+\varphi_{yy}}(|\varphi_{yy}|+|\varphi_{xx}|) \leqslant 1, |Re(F)_v| = \frac{1}{2+\varphi_{xx}+\varphi_{yy}} 2\varphi_{xy} \leqslant \frac{\sqrt{\varphi_{xx}\varphi_{yy}-1}}{1+\sqrt{\varphi_{xx}\varphi_{yy}}} \leqslant 1$。即 $|F'|$ 有界，根据刘维尔定理，F' 为常数，考虑 F' 的形式即得 $\varphi_{xx}, \varphi_{xy}, \varphi_{yy}$ 均为常数。

(2) 对于定义在 $\mathbb{R}^2$ 上的极小曲面 f，做 Levy 变换得到 (u,v) 参数，满足 (1) 中条件，即得 f_x, f_y 为常数，f 为线性函数。 □

3. (1) $\forall \overrightarrow{a} \in \mathbb{R}^3$, $l = \langle \overrightarrow{e_3}, \overrightarrow{a} \rangle$ 那么

$$dl = \langle d\overrightarrow{e_3}, \overrightarrow{a} \rangle = \langle \omega_{3i}\overrightarrow{e_i}, \overrightarrow{a} \rangle = -h_{ij}\langle \overrightarrow{e_i}, \overrightarrow{a} \rangle \omega_j.$$

于是

$$l_j = -h_{ij}\langle \overrightarrow{e_i}, \overrightarrow{a} \rangle.$$

再求一次共变导数

$$\begin{aligned} l_{ij}\omega_j &= dl_i + \omega_{ji}l_j \\ &= -d(-h_{ki}\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle) + \omega_{ji}(-h_{ij}\langle \overrightarrow{e_i}, \overrightarrow{a} \rangle) \\ &= -dh_{ki}\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle - h_{ki}\sum_{j=1}^{3}\omega_{kj}\langle \overrightarrow{e_j}, \overrightarrow{a} \rangle - \omega_{ji}h_{kj}\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle \\ &= -(dh_{ki} + h_{ji}\omega_{jk} + h_{kj}\omega_{ji})\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle - h_{ki}\omega_{k3}\langle \overrightarrow{e_3}, \overrightarrow{a} \rangle \\ &= -h_{ki,j}\omega_j\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle - h_{ki}h_{kj}\omega_j\langle \overrightarrow{e_3}, \overrightarrow{a} \rangle. \end{aligned}$$

于是

$$l_{ij} = -h_{ki,j}\omega_j\langle \overrightarrow{e_k}, \overrightarrow{a} \rangle - h_{ki}h_{kj}\langle \overrightarrow{e_3}, \overrightarrow{a} \rangle.$$

(2) 由于

$$\Delta_\Sigma l = \langle \Delta_\Sigma \overrightarrow{e_3}, \overrightarrow{a} \rangle.$$

结合 Codazzi 方程得到

$$\Delta_\Sigma l = l_{11} + l_{22} = -\langle (h_{11,k} + h_{22,k})\overrightarrow{e_k} + |A|^2\overrightarrow{e_3}, \overrightarrow{a} \rangle.$$

于是有

$$\Delta_\Sigma \overrightarrow{e_3} = -2H_{,k}\overrightarrow{e_k} - |A|^2\overrightarrow{e_3}.$$

(3) 取 $\overrightarrow{a} = (0,0,1)$, 那么 $l = \frac{1}{W}$. 带入第二问得到

$$\Delta_\Sigma l = -|A|^2\langle \overrightarrow{e_3}, \overrightarrow{a} \rangle = -|A|^2 l = \frac{|A|^2}{W}.$$

而

$$|A|^2 = \lambda_1^2 + \lambda_2^2 = -2\lambda_1\lambda_2 = -2K.$$

于是有

$$K = \frac{1}{2}W\Delta_\Sigma(\frac{1}{W}).$$

(4) 由 (1) 中计算得到

$$\begin{aligned}|\nabla(\frac{1}{W})|^2 &= (\lambda_1\langle\overrightarrow{e_1}, \overrightarrow{a}\rangle)^2 + (\lambda_1\langle\overrightarrow{e_1}, \overrightarrow{a}\rangle)^2 \\ &= \lambda_1^2(\langle\overrightarrow{e_1}, \overrightarrow{a}\rangle^2 + \langle\overrightarrow{e_2}, \overrightarrow{a}\rangle^2) \\ &= -K(1 - \langle\overrightarrow{e_3}, \overrightarrow{a}\rangle^2) \\ &= -K(1 - \frac{1}{W^2}).\end{aligned}$$

而

$$K = \Delta_\Sigma log(1 + \frac{1}{W}) = \frac{W}{W+1}\Delta_\Sigma(\frac{1}{W}) - \big(\frac{W}{W+1}\big)^2|\nabla(\frac{1}{W})|^2.$$

带入化简即可得到两式等价。 □

4. 利用 weierstrass 表示

$$x_1 = Re\frac{1}{2}\int f(1-g^2)dz \qquad x_2 = Re\frac{1}{2i}\int f(1+g^2)dz \qquad x_3 = Re\int fgdz$$

分别带入积分得到

(1) $x_1 = \frac{u}{2}$ $x_2 = \frac{v}{2}$ $x_3 = 0.$
(2) $x_1 = \frac{u}{2} - \frac{uv^2}{2} + \frac{u^3}{6}$ $x_2 = \frac{v}{2} + \frac{vu^2}{2} - \frac{v^3}{6}$ $x_3 = \frac{u^2-v^2}{2}.$
(3) $x_1 = \frac{u}{2} + \frac{u}{2}\frac{1}{u^2+v^2}$ $x_2 = \frac{v}{2} + \frac{v}{2}\frac{1}{u^2+v^2}$ $x_3 = \frac{1}{2}log(u^2+v^2).$
(4) $x_1 = -\frac{v}{2} + \frac{v}{2}\frac{1}{u^2+v^2}$ $x_2 = \frac{u}{2} - \frac{u}{2}\frac{1}{u^2+v^2}$ $x_3 = -arctan\frac{v}{u}.$
(5) $x_1 = \frac{1}{2}Re(arctanz)$ $x_2 = \frac{1}{2}Im(arctanz)$ $x_3 = \frac{1}{2}Re(arctanz^2).$

□

5. 由于 $g = diag(\lambda^2, \lambda^2)$, 带入 $\Delta_\Sigma = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}g^{ij}\partial_j) = \frac{1}{\lambda^2}(\partial_{ii} + \partial_{jj})$.

□